Nama : Tsaniya Faridah Maulida

NIM : 211331031

Kelas :1A

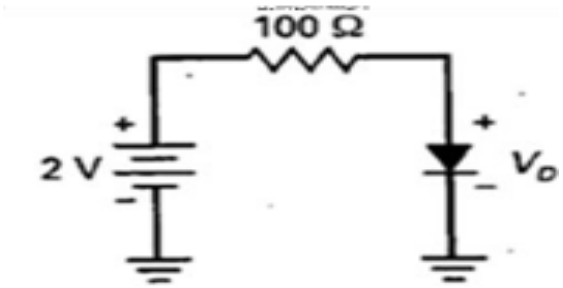
Mata Kuliah : Elektronika Analog

JURNAL GARIS BEBAN DIODA MENGGUNAKAN SIMULATOR

I. TUJUAN PERCOBAAN

1. Menggambarkan garis beban diode dan menunjukkan penggunaanya.
2. Menentukan titik kerja diode

II. GAMBAR RANGKAIAN



III. TABEL PERCOBAAN

Tabel 1.1 Dioda Silikon (IN40001)

Vs (V)	R (Ω)	Kondisi Dioda	Id(mA)	Vd(volt)
			ukur	ukur
1.5	470	terpasang	1,84	0,64
		lepas	0,00	1,50
		Hubungan singkat	3,19	0,00
3	2k Ω	terpasang	1,19	0,62
		lepas	0,00	3,00

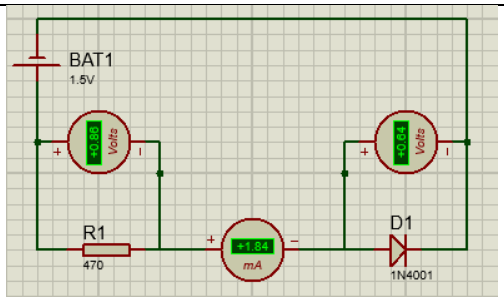
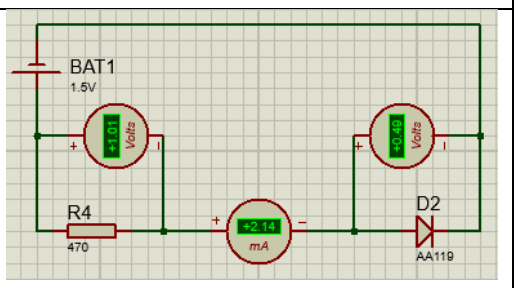
		Hubungan singkat	1,50	0,00

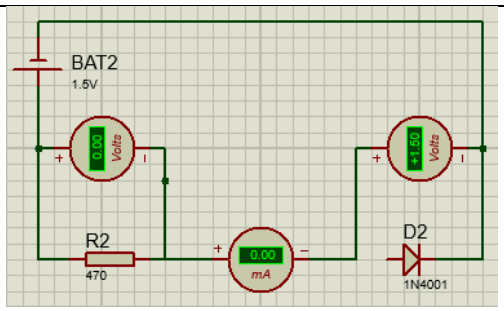
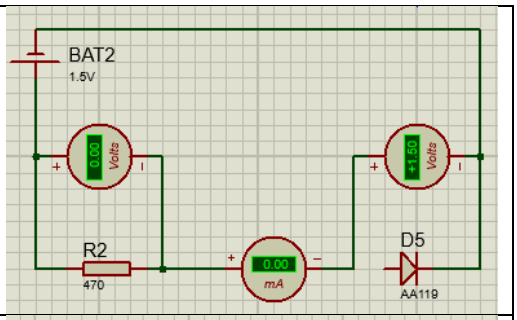
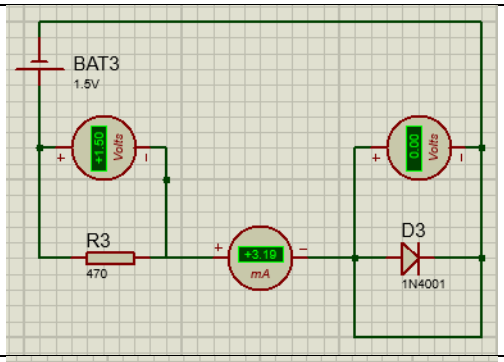
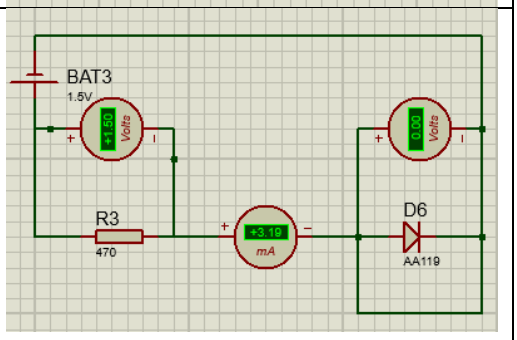
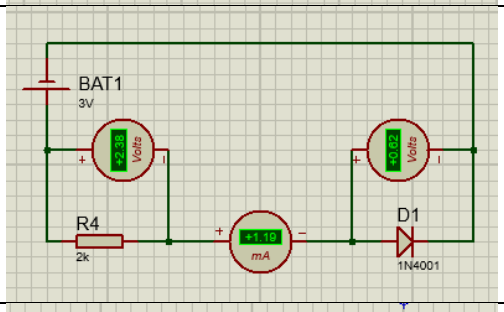
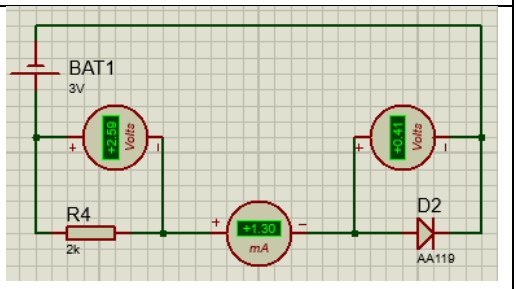
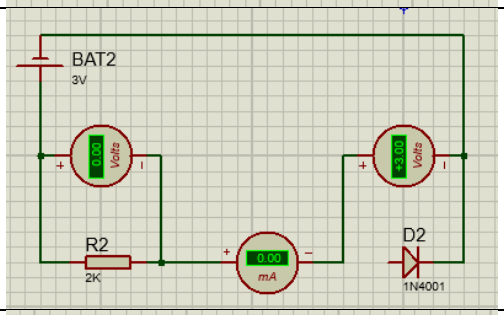
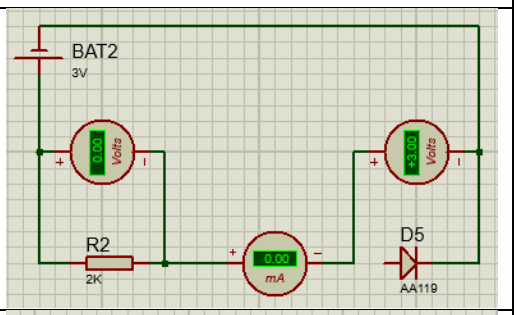
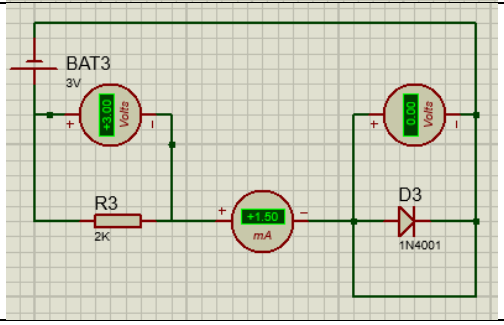
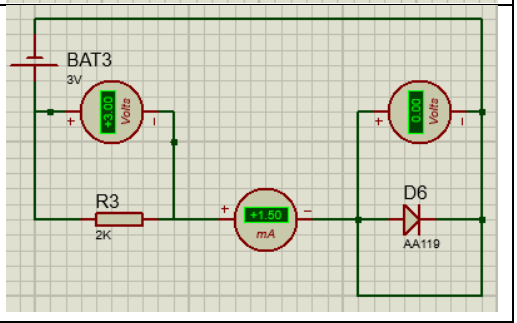
Table 1.2 Dioda Germanium (AA119)

Vs (V)	R (Ω)	Kondisi Dioda	Id(mA)	Vd(volt)
			ukur	ukur
1.5	470	terpasang	2,14	0,49
		lepas	0	1,50
		Hubungan singkat	3,19	0
3	2k Ω	terpasang	1,3	0,41
		lepas	0	3,00
		Hubungan singkat	1,5	0

IV. DOKUMENTASI HASIL PRAKTIKUM

Tabel 1.3 Foto praktikum diode silicon

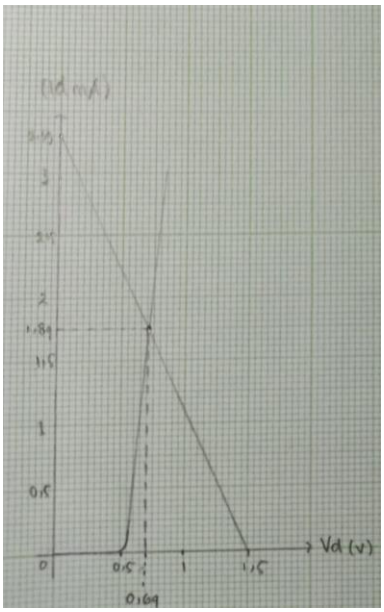
Vs (V)	R (Ω)	Kondisi Dioda	Silikon	Germanium
1,5	470	Terpasang		

		Lepas		
		Hubungan singkat		
3	2K	Terpasang		
		Lepas		
		Hubungan Singkat		

V. GRAFIK PERCOBAAN

Tabel 1.4 Data garis beban diode silicon Vs = 1,5 V

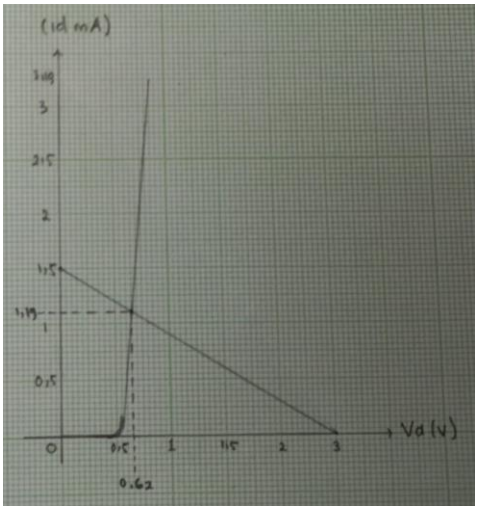
Keterangan	Arus DIoda (Id)	Tegangan Dioda (Vd)
Titik Operasi	1,84 mA	0,64 V
Titik Cut off	0 mA	1,5 V
Titik Saturasi	3,19 mA	0 V



Gambar 1.1 grafik garis beban diode silicon Vs = 1,5 V

Tabel 1.5 Data garis beban diode silikon Vs = 3 V

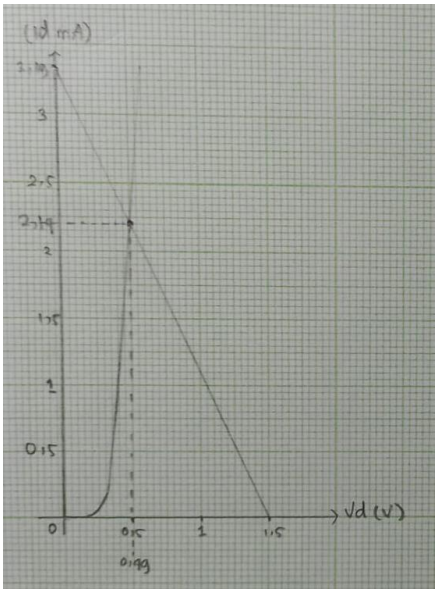
Keterangan	Arus DIoda (Id)	Tegangan Dioda (Vd)
Titik Operasi	1,19 mA	0,62 V
Titik Cut off	0 mA	3 V
Titik Saturasi	1,5 mA	0 V



Gambar 1.2 grafik garis beban diode silicon Vs = 3 V

Tabel 1.6 Data garis beban diode Germanium Vs = 1,5 V

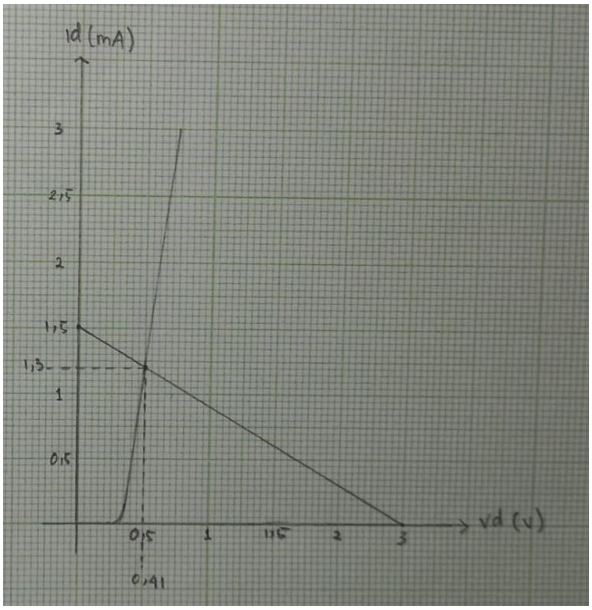
Keterangan	Arus DIoda (Id)	Tegangan Dioda (Vd)
Titik Operasi	2,14 mA	0,49 V
Titik Cut off	0 mA	1,5 V
Titik Saturasi	3,19 mA	0 V



Gambar 1.3 grafik garis beban diode Germanium Vs = 1,5 V

Tabel 1.7 Data garis beban diode Germanium Vs = 3 V

Keterangan	Arus DIoda (Id)	Tegangan Dioda (Vd)
Titik Operasi	1,3 mA	0,41 V
Titik Cut off	0 mA	3 V
Titik Saturasi	1,5 mA	0 V



Gambar 1.4 grafik garis beban diode Germanium Vs = 3 V